

GE23131-Programming using C-2024

Question 1

Correct Flag question Given an array of numbers, find the index of the smallest array element (the pivot), for which the sums of all elements to the left and to the right are equal. The array may not be reordered. Example arr=[1,2,3,4,6] · the sum of the first three elements, 1+2+3=6. The value of the last element is 6. · Using zero based indexing, arr[3]=4 is the pivot between the two subarrays. · The index of the pivot is 3.

Function Description Complete the function balancedSum in the editor below. Quiz navigation Show one page at a time Finish review 1 2 3

REC-CIS balancedSum has the following parameter(s): int arr[n]: an array of integers Returns: int: an integer representing the index of the pivot Constraints · $3 \leq n \leq 10$ · $1 \leq arr[i] \leq 2 \times 10$, where $0 \leq i < n$ · It is guaranteed that a solution always exists. Input Format for Custom Testing Input from stdin will be processed as follows and passed to the function. The first line contains an integer n, the size of the array arr. Each of the next n lines contains an integer, arr[i], where $0 \leq i < n$.

Sample Case 0 Sample Input 0 STDIN Function Parameters -----

----- 4 → arr[] size n = 4 1 → arr = [1, 2, 3, 3] 5 4 REC-CIS 2 3 3 Sample Output 0 2 Explanation 0 · The sum of the first two elements, 1+2=3. The value of the last element is 3. · Using zero based indexing, arr[2]=3 is the pivot between the two subarrays. · The index of the pivot is 2.

Sample Case 1 Sample Input 1 STDIN Function Parameters -----

----- 3 → arr[] size n = 3 1 → arr = [1, 2, 1] 2 1 Sample Output 1 REC-CIS 1 Explanation 1 · The first and last elements are equal to 1. · Using zero based indexing, arr[1]=2 is the pivot between the two subarrays. · The index of the pivot is 1. Answer: (penalty regime: 0 %)

```

1  int balancedSum(int arr_count, int* arr)
2  {
3      int totalsum=0;
4      for(int i=0;i<arr_count;i++)
5      {
6          totalsum+=arr[i];
7      }
8      int leftsum=0;
9      for(int i=0;i<arr_count;i++)
10     {
11         int rightsum=totalsum-leftsum-arr[i];
12         if(leftsum==rightsum)
13         {
14             return i;
15         }
16         leftsum+=arr[i];
17     }
18     return 1;
19 }
20

```

	Test	Expected	Got	
✓	int arr[] = {1,2,3,3}; printf("%d", balancedSum(4, arr))	2	2	✓

Passed all tests! ✓

Question 2

Correct Flag question Test Expected Got int arr[] = {1,2,3,3};
printf("%d", balancedSum(4, arr)) 2 2 Passed all tests! Calculate the sum of an array of integers. Example numbers = [3, 13, 4, 11, 9] The sum is 3 + 13 + 4 + 11 + 9 = 40. Function Description Complete the function arraySum in the editor below. arraySum has the following parameter(s): int numbers[n]: an array of integers Returns int: integer sum of the numbers array Constraints REC-CIS $1 \leq n \leq 10$ $1 \leq \text{numbers}[i] \leq 10$ Input Format for Custom Testing Input from stdin will be processed as follows and passed to the function. The first line contains an integer n, the size of the array numbers. Each of the next n lines contains an integer numbers[i] where $0 \leq i < n$. Sample Case 0 Sample Input 0 STDIN Function ----- 5 → numbers[] size n = 5 1 → numbers = [1, 2, 3, 4, 5] 2 3 4 5 Sample Output 0 15 4 4 REC-CIS Explanation 0 1 + 2 + 3 + 4 + 5 = 15. Sample Case 1 Sample Input 1 STDIN Function ----- 2 → numbers[] size n = 2 12 → numbers = [12, 12] 12 Sample Output 1 24 Explanation 1 12 + 12 = 24. Answer: (penalty regime: 0 %)

```
1 int arraySum(int numbers_count, int *numbers)
2 {
3     int sum=0;
4     for(int i=0;i<numbers_count;i++)
5     {
6         sum=sum+numbers[i];
7     }
8     return sum;
9 }
10
```

	Test	Expected	Got	
✓	int arr[] = {1,2,3,4,5}; printf("%d", arraySum(5, arr))	15	15	✓

Passed all tests! ✓

Question 3

Correct Flag question { sum=sum+numbers[i]; } return sum; } Test Expected Got int arr[] = {1,2,3,4,5}; printf("%d", arraySum(5, arr)) 15 15 Passed all tests! Given an array of n integers, rearrange them so that the sum of the absolute differences of all adjacent elements is minimized. Then, compute the sum of those absolute differences. Example n = 5 arr = [1, 3, 3, 2, 4] If the list is rearranged as arr' = [1, 2, 3, 3, 4], the absolute differences are $|1 - 2| = 1$, $|2 - 3| = 1$, $|3 - 3| = 0$, $|3 - 4| = 1$. The sum of those differences is $1 + 1 + 0 + 1 = 3$. Function Description Complete the function minDiff in the editor below. minDiff has the following parameter: arr: an integer array Returns: int: the sum of the absolute differences of adjacent elements Constraints $2 \leq n \leq 105$ $0 \leq arr[i] \leq 109$, where $0 \leq i < n$ Input Format For Custom Testing The first line of input contains an integer, n, the size of arr. Each of the 6 7 8 9 10 REC-CIS following n lines contains an integer that describes arr[i] (where $0 \leq i < n$) . Sample Case 0 Sample Input For Custom Testing STDIN Function ----- 5 $\rightarrow$ arr[] size n = 5 5 $\rightarrow$ arr[] = [5, 1, 3, 7, 3] 1 3 7 3 Sample Output 6 Explanation n = 5 arr = [5, 1, 3, 7, 3] If arr is rearranged as arr' = [1, 3, 3, 5, 7], the differences are minimized. The final answer is $|1 - 3| + |3 - 3| + |3 - 5| + |5 - 7| = 6$. Sample Case 1 Sample Input For Custom Testing STDIN Function ----- 2 $\rightarrow$ arr[] size n = 2 3 $\rightarrow$ arr[] = [3, 2] 2 Sample Output 1 Explanation n = 2 arr = [3, 2] There is no need to rearrange because there are only two elements. The final answer is $|3 - 2| = 1$. Answer: (penalty regime: 0 %)

```

1  #include<stdlib.h>
2  int compare(const void*a,const void*b)
3  {
4      return (*(int*)a-*(int*)b);
5  }
6  int minDiff(int arr_count,int *arr)
7  {
8      qsort(arr,arr_count,sizeof(int),compare);
9      int totaldiff=0;
10     for(int i=1;i<arr_count;i++)
11     {
12         totaldiff+=abs(arr[i]-arr[i-1]);
13     }
14     return totaldiff;
15 }

```

	Test	Expected	Got	
✓	int arr[] = {5, 1, 3, 7, 3}; printf("%d", minDiff(5, arr))	6	6	✓

Passed all tests! ✓